Computational Methods #7

CH7 Exercise 2

The velocity distribution of a thermalized gas at a temperature $\tau = k_B T$ is,

$$\rho(v)\,dv = \left(\frac{m}{2\pi\tau}\right)^{3/2} 4\pi v^2 e^{-mv^2/2\tau} dv$$

The first task is to find a natural velocity scale for the problem v_0. Given the energy τ and mass m the natural velocity scale is $v_0^2 = \tau/m$. Defining the non-dimensionalized velocity $u = v/v_0$,

$$\rho(u)\,du = \sqrt{\frac{2}{\pi}} u^2 e^{-u^2/2} du$$

Where $\rho(u(v)) = v_0 \rho(v)$. This can be written in terms of the Gamma distribution

$$\rho_\Gamma(x|\alpha = 3/2, \beta = 2) = \frac{1}{\sqrt{2\pi}} \sqrt{x} e^{-x/2}$$

Using the transformation $x = u^2$ we find that this is the same as the velocity distribution of a thermal gas.

$$\begin{aligned}
\rho_\Gamma(x)\,dx &= \frac{1}{\sqrt{2\pi}} \sqrt{x} e^{-x/2} dx \\
&= \frac{1}{\sqrt{2\pi}} u e^{-u^2/2} du^2 \\
&= \sqrt{\frac{2}{\pi}} u^2 e^{-u^2/2} du \\
&= \rho(u)\,du
\end{aligned}$$

The procedure for generating the histogram is

- Sample x from a Gamma distribution
- Transform the sampled x to $u = \sqrt{x}$
- Add u to a histogram

```
Hist1D hx ("X", NumBins, umin, umax);
  int NumSamples = 1000000;
  for(int i=0; i<NumSamples; i++){
    double x = rx(e);
    double u = sqrt(x);
    hx.accumulate(u);
  }
```

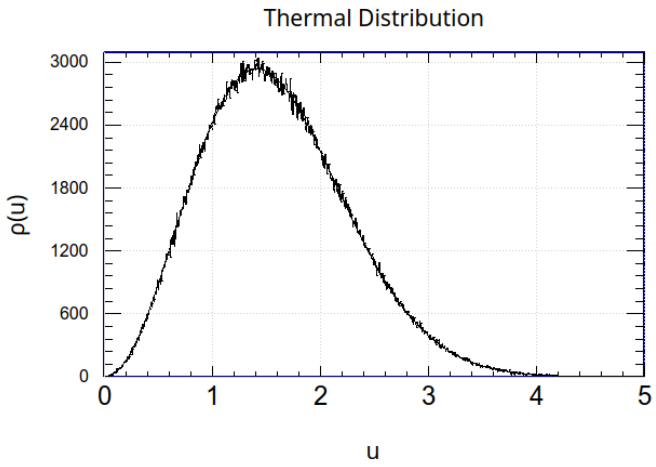


Figure 1: Velocity distribution of a thermal gas.

This procedure generates the histogram Fig[1].

Exercise 3

The same histogram is generated using a Markov Chain Monte Carlo algorithm.

- From an initial velocity $\vec{v} = \vec{0}$
- Nudge the elements v_x, v_y, v_z using a Gaussian distributed nudge with step size σ.
- If $P(\vec{v}_{\text{new}}) > P(\vec{v}_{\text{old}})$ then move to $\vec{v}_{\text{new}}$ otherwise move to $\vec{v}_{\text{new}}$ with a probability $P(\vec{v}_{\text{new}})/P(\vec{v}_{\text{old}})$
- Add $v = |\vec{v}_{\text{new}}|$ to the histogram

```
//MCMC loop
for(int i=0; i<NStates; i++){

  //propose a new velocity
  std::vector<double> v_new = v;
      //random step
  for(int j=0; j<3; j++) {v_new[j] += gauss(e);}

  double v_new_norm = v_new[0]*v_new[0]
                    + v_new[1]*v_new[1]
                    + v_new[2]*v_new[2];
  double Prob_v_new = rhoNorm(v_new_norm);

  //accept or reject the new velocity
  if(flat(e) < std::min(1.0,Prob_v_new/Prob_v)) {
      v = v_new;
      v_norm = v_new_norm; Prob_v = Prob_v_new; accepted+=1;
      }

  //add new velocity to histogram
  hx.accumulate(sqrt(v_norm));
}
```

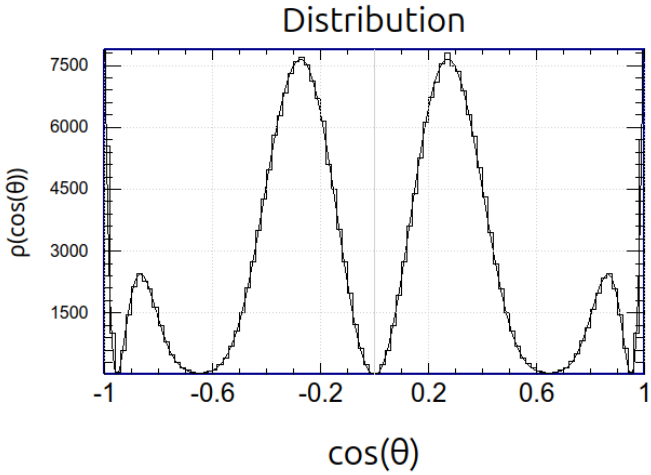
Exercise 6

The distribution in this problem is creating using the rejection method.

- Sample x and y from uniform distributions
- if $y < \rho(x)$ add x to the histogram

```
Hist1D hx ("X", NumBins, umin, umax);
int NumSamples = 1000000;
for(int i=0; i<NumSamples; i++){
  double x = rx(e);
  double y = ry(e);

  if(y < rho(x)) {hx.accumulate(x);NAccepted+=1;}
}
```



Exercise 8

The Hydrogen molecule ion Hamiltonian is given by

$$H = -\frac{1}{2}\nabla^2 - \frac{1}{\left|\vec{r} - \vec{d}/2\right|} - \frac{1}{\left|\vec{r} + \vec{d}/2\right|}$$

where $\vec{d} = d\hat{z}$. (Not sure why in the corrected version the kinetic energy operator is $-\frac{1}{2}\nabla_1^2 - \frac{1}{2}\nabla_2^2$. This would imply that there are two different electros with positions $\vec{r}_1$ and $\vec{r}_2$ each of which is operated on by $\partial/\partial\vec{r}_1$ or $\partial/\partial\vec{r}_2$ which is not the given setup so I will continue with the old Hamiltonian). The approximate solutions to the problem are symmetric and anti-symmetric linear combinations of atomic orbitals,

$$|\psi_{s,a}\rangle = N_{s,a}\left(|\phi_1\rangle \pm |\phi_2\rangle\right)$$

To find these normalization factors $N_{s,a}$ set the norm of the wave functions to 1.

$$\begin{aligned}\langle\psi_{s,a}|\psi_{s,a}\rangle &= 2N_{s,a}^2\left(\langle\phi_1|\phi_1\rangle \pm \langle\phi_1|\phi_2\rangle\right)\\ &= 2N_{s,a}^2\left(1 \pm \langle\phi_1|\phi_2\rangle\right)\end{aligned}$$

Evaluating the inner product $\langle\phi_1|\phi_2\rangle$ by Monte Carlo integration and using a change of variable $\vec{r}' = \vec{r} + \vec{d}/2$ to place one of the nuclei at the origin of integration,

$$\begin{aligned}\langle\phi_1|\phi_2\rangle &= \frac{1}{\pi}\int dV e^{-|\vec{r}+\vec{d}/2|-|\vec{r}-\vec{d}/2|}\\ &= \frac{1}{\pi}\int dV e^{-r-|\vec{r}-\vec{d}|}\\ &= \frac{1}{\pi}\int d\phi\, d\cos\theta\, dr\, r^2 e^{-r-\sqrt{r^2+d^2-2rd\cos\theta}}\\ &= 2\int_{-1}^{1} du \int_0^{\infty} dr\, r^2 e^{-r-\sqrt{r^2+d^2-2dru}}\end{aligned}$$

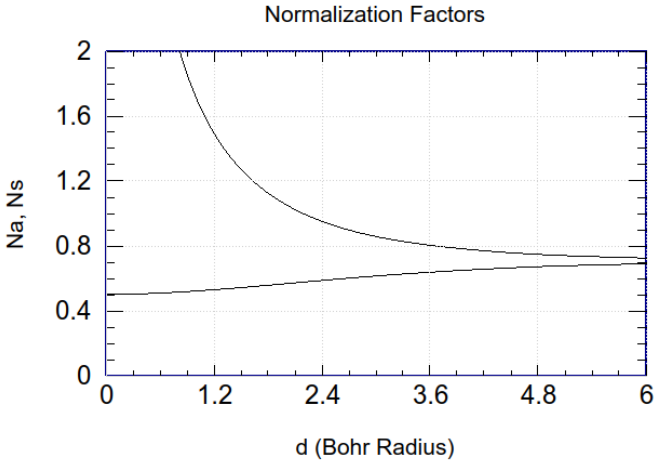


Figure 2: Normalization Factors

Using the exponential e^{-r} as a weight function we can evaluate this integral using Monte Carlo integration.

As seen in Fig[2] as the inter-atomic distance gets bigger the basis functions become more orthogonal and therefore N_a and N_b both approach $1/\sqrt{2}$ as expected.

The total energy is calculated as the matrix element,

$$\langle\psi_{s,a}|\, H\, |\psi_{s,a}\rangle = N_{s,a}^2\left(\langle\phi_1|\, H\, |\phi_1\rangle \pm 2\,\langle\phi_1|\, H\, |\phi_2\rangle + \langle\phi_2|\, H\, |\phi_2\rangle\right)$$

The diagonal matrix elements of H are,

$$\begin{aligned}
\langle\phi_1|\, H\, |\phi_1\rangle &= \frac{1}{\pi}\int dV\,\left(-\frac{1}{2}\phi_1\nabla^2\phi_1 - \frac{\phi_1^2}{\left|\vec{r}-\vec{d}/2\right|} - \frac{\phi_1^2}{\left|\vec{r}+\vec{d}/2\right|}\right)\\
&= \frac{1}{\pi}\int dV\, e^{-2|\vec{r}-\vec{d}/2|}\left(-\frac{\left|\vec{r}-\vec{d}/2\right|^2}{2} - \frac{1}{\left|\vec{r}-\vec{d}/2\right|} - \frac{1}{\left|\vec{r}+\vec{d}/2\right|}\right)\\
&= \frac{1}{\pi}\int d\Omega\int dr\, r^2 e^{-2r}\left(-\frac{r^2}{2} - \frac{1}{r} - \frac{1}{\left|\vec{r}+\vec{d}\right|}\right)\\
&= 2\int_{-1}^{1} du\int dr\, e^{-2r}\left(-\frac{r^4}{2} - r^3 - \frac{r^2}{\sqrt{r^2+d^2+2rdu}}\right)
\end{aligned}$$

The off diagonal matrix elements of H are,

$$\begin{aligned}
\langle\phi_2|\, H\, |\phi_1\rangle &= \frac{1}{\pi}\int dV\left(-\frac{1}{2}\phi_2\nabla^2\phi_1 - \frac{\phi_1\phi_2}{\left|\vec{r}-\vec{d}/2\right|} - \frac{\phi_1\phi_2}{\left|\vec{r}+\vec{d}/2\right|}\right)\\
&= \frac{1}{\pi}\int d\Omega\int dr\, e^{-r}e^{-|\vec{r}+\vec{d}|}\left(-\frac{r^4}{2} - r^3 - \frac{2}{\left|\vec{r}+\vec{d}\right|}\right)\\
&= 2\int_{-1}^{1} du\int dr\, e^{-r}e^{-\sqrt{r^2+d^2+2rdu}}\left(-\frac{r^4}{2} - r^3 - \frac{r^2}{\sqrt{r^2+d^2+2rdu}}\right)
\end{aligned}$$

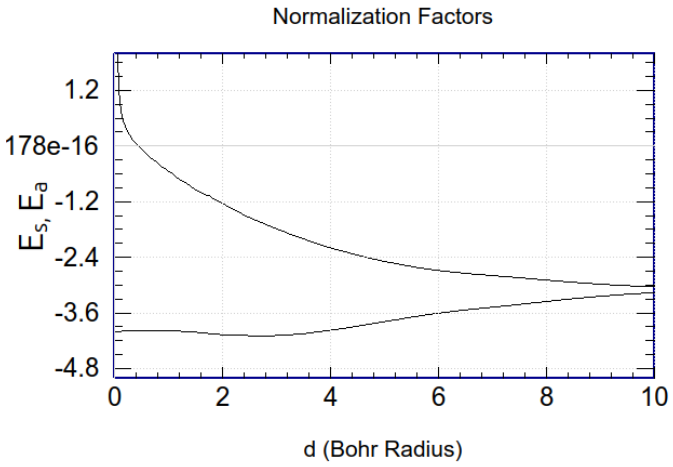


Figure 3: Normalization Factors

As can be seen in Fig[3] for large distances d between the nuclei the two states converge to the same energy as the interaction energy $\langle\phi_1|\, H\, |\phi_2\rangle$ vanishes.

Exercise 12

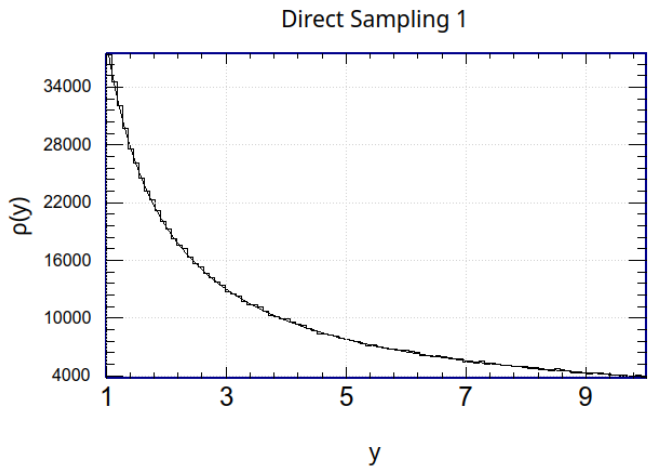
Given a transformation function f that transforms the uniform variate x to the non-uniform variate y through $y = f(x)$. We want to find the distribution function $\rho(y)$ that the variate y is sampled from.

1. For the transformation function $f(x) = x_{\min}\left(x_{\max}/x_{\min}\right)^x$ where x is uniformly sampled from $[0,1]$. The first step is to find the inverse transformation $x = f^{-1}(y)$.

$$x = \frac{\ln\left(y/x_{\min}\right)}{\ln\left(x_{\max}/x_{\min}\right)}$$

The probability density function is then given by,

$$\begin{aligned}\rho(y) &= \frac{df^{-1}}{dy} \\ &= \frac{1/y}{\ln(x_{\max}/x_{\min})}\end{aligned}$$



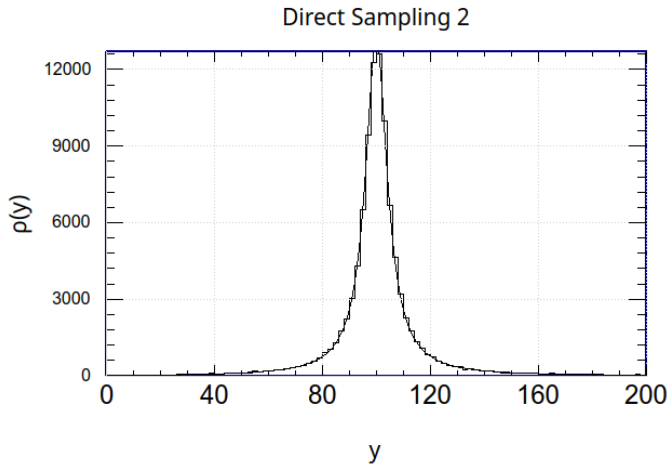
2. For the transformation function $f(x) = \sqrt{M^2 + M\Gamma\tan(M\Gamma x)}$ where x is uniformly sampled from $\left[-\tan^{-1}(M/\Gamma)/M\Gamma, \pi/2M\Gamma\right]$ such that the range of y is from $[0, \infty)$. The first step is to invert the transformation

$$x = \frac{1}{M\Gamma}\tan^{-1}\left(\frac{y^2 - M^2}{M\Gamma}\right)$$

The probability density function is then given by,

$$\begin{aligned}\rho(y) &= \frac{1}{x_{\max} - x_{\min}}\frac{df^{-1}}{dy} \\ &= \frac{1}{x_{\max} - x_{\min}}\frac{2y}{(M\Gamma)^2 + (y^2 - M^2)^2}\end{aligned}$$

Where the factor $\frac{1}{x_{\max} - x_{\min}}$ is $\rho_x(x)$ which is not 1 since I have chosen a range different from $[0, 1]$ to have a nice output range.


Direct Sampling 2
12000
9000
6000
3000
0
ρ(y)
0
40
80
120
160
200
y